# Grid Layout Properties

- ✓ **CSS Grid Layout** is a **two-dimensional layout system**. It allows you to arrange content in **rows and columns** easily.
- ✓ It's perfect for designing **web page structures**, **galleries**, **dashboards**, or **complex layouts** without relying on floats or flexbox hacks.

| Property | Description | Possible Values | Default Value | Syntax / Example |
|---|---|---|---|---|
| **grid-template-columns** | Defines the number and width of columns in a grid layout. | Fixed units (px, em, %) | none | grid-template-columns: 200px 1fr 2fr;<br>The **fr** unit means **"fraction of the available space."** |
| **grid-template-rows** | Defines the number and height of rows in a grid layout. | Fixed units (px, em, %), | none | grid-template-rows: 100px auto 100px; |
| **gap (grid-gap)** | Defines the space between rows and columns. | Any CSS length (px, em, %) | 0 | gap: 10px; or gap: 20px 40px; |
| **grid-column** | Specifies the horizontal position of an item within the grid columns (start / end lines). | <start-line> / <end-line> | auto | grid-column: 1 / 3; (spans across 2 columns) |
| **grid-row** | Specifies the vertical position of an item within the grid rows (start / end lines). | <start-line> / <end-line> | auto | grid-row: 1 / 2; (spans across 1 row) |

## Example:

```
<head>
<style>
/* ===== Example 1: Fixed + Flexible Columns ===== */
.grid1 {
  display: grid;
  grid-template-columns: 200px 1fr 2fr; /* 3 columns: fixed + flexible */
  grid-template-rows: 1fr 2fr; /* 2 flexible rows (1:2 ratio) */
  gap: 10px;
  background-color: lightgray;
  padding: 10px;
  margin-bottom: 40px;
}
/* ===== Example 2: Equal Columns ===== */
```

```
.grid2 {
  display: grid;
  grid-template-columns: repeat(4, 1fr); /* 4 equal columns */
  grid-template-rows: 100px;
  height:150px;
  gap: 10px;
  background-color: lightblue;
  padding: 10px;
}
.item {
  background-color: steelblue;
  color: white;
  text-align: center;
  padding: 20px;
  border-radius: 10px;
  font-size: 18px;
}
</style>
</head>
<body>
<div class="grid1">
  <div class="item">200px (Fixed)</div>
  <div class="item">1fr (Flexible)</div>
  <div class="item">2fr (Twice as Wide)</div>
</div>
<div class="grid2">
  <div class="item">Item 1</div>
  <div class="item">Item 2</div>
  <div class="item">Item 3</div>
  <div class="item">Item 4</div>
</div>
</body>
```

